Physics 603
Spring 2025
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Homework 4

Due Friday, February 28

The components of an angular momentum $\mathbf{J}$ (which can be orbital, spin, or total) can be organized into a spin-1 tensor operator as J_{+1}, J_0, J_{-1}, where $J_0 = J_z$, and

$$J_{\pm 1} = \mp \frac{1}{\sqrt{2}}(J_x \pm iJ_y) = \mp J^{\pm} \,.$$

Note in particular that $J_{+1} = -J^+$. For this homework, please assume $\hbar = 1$.

Problems:

1. Problem 4.B from the textbook. You do not have to answer the last question of the problem.

2. Let $\mathbf{J} = \mathbf{L} + \mathbf{S}$, where $\mathbf{L}$ is the orbital angular momentum, and $\mathbf{S}$ is spin. Denote the standard basis states of the spin-J irrep of $\mathbf{J}$ as $|J, M\rangle$. Please assume $\hbar = 1$.
   (a) Suppose that
   $$\langle 1, 1|S_z|1, 1\rangle = A$$
   is known. Compute
   $$\langle 1, M|S_z|1, M\rangle$$
   for the two remaining values of M.
   (b) Compute
   $$\langle 1, M|S^-|1, M + 1\rangle$$
   for all possible M, in terms of the same constant A as in (a).
   (c) Same for
   $$\langle 1, M|S^+|1, M - 1\rangle \,.$$
   (d) Because J_z and $J^{\pm}$ acting on $|1, M\rangle$ each produce another state of the same irrep (and we know what state that is), one can use the results of (b) and (c) to compute, in terms of A, the matrix element
   $$\langle 1, M|\mathbf{S} \cdot \mathbf{J}|1, M\rangle \,.$$
   Compute it (in terms of A) for all M, using the observation that
   $$\mathbf{S} \cdot \mathbf{J} = S^+J^- + S^-J^+ + S_zJ_z \,.$$